Qns	Answer	Marks
8(a)	$hc / \lambda = \Phi + E_{MAX}$ AND hc = gradient	C1
	gradient = e.g. $[(0.40 - 0.20) \times 1.60 \times 10^{-19}] / [(2.25 - 2.09) \times 10^{6}]$ (*working needed*) $(= 2.0 \times 10^{-25})$	M1
	$h = (2.0 \times 10^{-25}) / (3.00 \times 10^{8}) = 6.7 \times 10^{-34}$ J s (*both working and answer needed*)	A1
8(b)	straight line with same gradient as the original AND straight line with x-axis intercept greater than 1.93×10^{6} m^{-1}	B1
8(c)	(for $E_{MAX} = 0$,) $1 / \lambda_0 = 1.93 \times 10^{6}$ (m^{-1}) $f_0 = (3.00 \times 10^{8})(1.93 \times 10^{6})$ $= 5.8 \times 10^{14}$ Hz	C1
	$\Phi = hf_0 = (6.63 \times 10^{-34})(5.8 \times 10^{14})$ $= 3.9 \times 10^{-19}$ J = 2.4 eV	M1
	so potassium	A1
8(d)	more photons (per unit time) so (rate of emission) increases	A1